The scalar glueball on center-vortex-removed SU(2) lattice ensembles: a data note

Turaq Rayn-McLean

October 30, 2026 (draft, not submitted)

Abstract

We measure the zero-momentum 0^{++} glueball correlator on SU(2) Wilson-action ensembles at $\beta = 2.3$, 2.4 and 2.5 in three readings of the same configurations: the original links, the center-projected Z_2 links in direct maximal center gauge, and the vortex-removed links of de Forcrand and D'Elia. A literature search found no published glueball measurement on vortex-removed ensembles; this is new as far as we found, not proven new. At $\beta = 2.4$ the result is clean: the vortex-removed string tension is zero within errors, and the vortex-removed 0^{++} correlator is a single exponential with mass $2.40(8)\sqrt{\sigma}$ (original string tension), 0.71(4) of the original glueball mass, unchanged between 12^4 and 16^4 (ratio 0.98(4)) and below the free two-gluon level. At $\beta = 2.5$ the vortex-removed mass is also volume independent ($20^4/16^4$ ratio 0.94(6)), but our pre-registered premise check fails there: a short-time string-tension estimator leaves about 0.21 of the original string tension on the vortex-removed ensemble, while longer-time estimators are consistent with zero. A paired study with the best of 16 gauge copies does not remove that residual (0.225(10)), lowers the center-projected string tension to 0.73 of the full one, and shifts the vortex-removed mass by 2.3σ. At $\beta = 2.3$ the vortex-removed effective mass does not plateau on the larger box. Applying our pre-registered rules exactly, the combined outcome is inconclusive. We publish the measurements, the failure modes, data and code.

1. Limitations, stated first

- Monte Carlo evidence on small lattices: boxes from 8^4 to 20^4, at most two volumes per coupling in each volume test.
- SU(2) only, Wilson action, three couplings ($\beta = 2.3, 2.4, 2.5$), no continuum extrapolation. Nothing here bears on the continuum mass-gap problem.
- The vortex-removed (VR) ensemble is gauge dependent: it is defined through direct maximal center gauge (MCG), which has many local maxima (Gribov copies). The main runs use one gauge copy per configuration; the $\beta = 2.5$ follow-up shows that the VR mass and the center-projected (CP) string tension move with the copy.
- VR configurations are not generated by a local reflection-positive action, so VR correlators need not be positive sums of exponentials; "mass" means the decay rate read by our estimator.
- Estimator caveats: effective masses are resolved only to $t \approx 3$; the volume tests use a fixed plateau rule on $m(1), m(2)$; the string-tension premise check uses $V(R) = \ln W(R,3)/W(R,4)$, which at $\beta = 2.5$ probes short physical times. Our $\beta = 2.5$ unit $a\sqrt{\sigma} = 0.2072(10)$ is about 10% above the literature value with this estimator. Original glueball masses come out about 10% below continuum values [9, 10].
- Not blind: the original-ensemble glueball mass was known in advance; the first series' estimators were chosen after seeing exploratory 8^4 and 12^4 data; one deviation (GEVP t_0) was made after

testing on already-seen data; the long-time estimators of sec. 4.3 and the gauge-copy method of sec. 4.4 were chosen after the $\beta = 2.5$ premise failure was known. Each series did fix its rules and wall-clock stopping time before its own production data existed.

2. Background: what is known and what is new

De Forcrand and D'Elia [1] removed center vortices from SU(2) configurations by multiplying each link by its center projection in direct MCG, and found that confinement is lost (Creutz ratios tend to zero, large-T slopes do not grow with R) at $\beta = 2.4$ on 16^4. Pepe and de Forcrand [2] found the same at $\beta = 2.3$ to 2.5 with Laplacian center gauge. Direct MCG and center dominance go back to Del Debbio et al. [3, 4]. Vortex removal has since been used for gluon and ghost propagators [5, 6, 7] and for the light hadron spectrum and chiral symmetry in SU(3) [8], where it restores chiral symmetry. Glueballs after Abelian projection were measured in [11, 12]. Langfeld and Schafke [13] measured 0^+ and 2^+ screening masses on thick-vortex ensembles obtained by gauge-invariant cooling, which is a different reduction from the one used here. The dependence of center-projected observables on the MCG gauge copy is documented in [14, 15]: better maxima lower the projected string tension, and the effect weakens on larger volumes.

New here, as far as our search found (arXiv, INSPIRE including the papers citing [1], 1999 to October 2026): (a) the 0^{++} glueball correlator on VR ensembles and on thin Z_2 CP ensembles, with volume tests; (b) the observation that at $\beta = 2.5$ a short-time estimator leaves a residual VR string tension of about 0.2 while long-time estimators do not; (c) a paired gauge-copy study of that residual and of the VR mass. Item (b) is consistent with [1, 2], which used long-distance observables.

3. Method

Ensembles. SU(2) Wilson action, heat bath with four over-relaxation steps per compound sweep, hot start, 300 thermalization sweeps. Three series, each with independent streams stopped by a wall clock fixed before any of its data were analysed:

- Series 1: $\beta = 2.3$ on 8^4, 12^4, 16^4 and $\beta = 2.4$ on 12^4, 16^4, $T = L$, a measurement every 2 sweeps (the 8^4 and part of the 12^4 statistics come from earlier exploratory streams with the same action and gauge fixing).
- Series 2: $\beta = 2.3$ on $12^3 \times 32$ and $16^3 \times 32$; $\beta = 2.5$ on 16^4 and 20^4; a measurement every 2 sweeps.
- Series 3 (gauge-copy study): $\beta = 2.5$ on 16^4 (8 streams) and 20^4 (2 streams), $\beta = 2.3$ on $12^3 \times 32$ (2 streams, control); a measurement every 10 sweeps.

Center projection and vortex removal. Direct MCG maximizes $\sum_{x,\mu} |\mathrm{tr}\, U_\mu(x)|^2$ by local exact maximization with over-relaxation ($\omega = 1.85$) until the relative change is below 10^{-8}. With $Z_\mu(x) = \mathrm{sign}\,\mathrm{tr}\, U_\mu(x)$, the CP links are $Z_\mu(x)$ and the VR links are $Z_\mu(x)U_\mu(x)$ [1]. On every plaquette of every configuration we assert $P_{\mathrm{orig}} = Z_P P_{\mathrm{VR}}$, which fails if the gauge fixing is not a gauge transformation. Series 1 and 2 use one gauge copy (over-relaxation started from the configuration as generated). Series 3 fixes every configuration twice: the same single copy ("single"), and the best functional among 16

candidates, the single copy plus 15 Haar-random gauge transforms each relaxed to convergence ("best of 16"). Both are measured on the same configuration, so differences are paired.

Glueball. Zero-momentum sums of spatial Wilson loops (1×1, 2×2, 3×3, 4×4, 1×2, 2×3, 3×4) at three APE smearing levels ($\alpha = 0.5$; $(0, 2, 8)$ steps at $\beta = 2.3$, $(0, 4, 15)$ at 2.4, $(0, 8, 31)$ at 2.5, scaled with $(a\sqrt{\sigma}\,)^{-2}$): basis A, 21 operators (CP: the 7 shapes unsmeared). Vacuum-subtracted correlators averaged over all time slices, normalized by $\sqrt{\mathrm{diag}\, C(0)}$, ground vector from the GEVP at $(t_0, t) = (0, 1)$, effective masses $m(t) = \ln C(t)/C(t+1)$. Plateau rule, fixed in advance: if $m(2)$ has relative error below 30% and $|m(1) - m(2)| < 1.5\sigma$ (jackknife of the difference), quote $m(1)$ ("plateau"); if $m(2)$ is that precise but differs, quote $m(2)$ ("late"); otherwise quote $m(1)$ as an "upper bound"; a quoted value with relative error above 30% is "noisy".

String tension. Estimator E1: $V(R) = \ln W(R, 3)/W(R, 4)$, fitted with $V_0 + \sigma R - e/R$, e free, $R = 2, \ldots, L/2$, with spatially smeared Wilson loops. Series 3 adds E1′ ($R \geq 4$), E2 ($T = 5 \to 6$ and $6 \to 7$) and Creutz ratios $\chi(R, R)$. Ratios $\sigma_X/\sigma_{\mathrm{ORIG}}$ use the same estimator on the same configurations.

Errors. Delete-one-block jackknife, block length at least twice the largest integrated autocorrelation time (Madras-Sokal window, $c = 6$), 8 to 20 blocks per stream; a bin-size scan is reported in the data. Thermalization check: first quarter dropped if its plaquette differs from the rest by more than 3σ (one stream in series 1).

Pre-registered tests. Premise C0: $\sigma_{\mathrm{VR}}/\sigma_{\mathrm{ORIG}} < 0.15$ and $\sigma_{\mathrm{CP}}/\sigma_{\mathrm{ORIG}} \in [0.6, 1.4]$ with E1. Volume test at each coupling: $\rho = m(L_{\mathrm{large}})/m(L_{\mathrm{small}})$ in lattice units; "gap" if ρ is within 2σ of 1 and more than 2σ from the box ratio $L_{\mathrm{small}}/L_{\mathrm{large}}$, "gapless" the reverse, otherwise inconclusive; interpreted only if the original-ensemble control reads "gap".

4. Results

4.1 Statistics and checks

Plaquettes: 0.60228 ($\beta = 2.3$), 0.63001 (2.4), 0.6519 to 0.6520 (2.5), all within 3×10^{-4} of the literature values. Integrated autocorrelation times are 0.5 to 1.6 measurements (one short stream 3.0). No assertion fired in any stream. MCG functional: 0.753 (2.3), 0.773 (2.4), 0.791 (2.5, single copy). P-vortex density: 0.090, 0.055, 0.032.

4.2 Masses

Table 1. Basis A, lattice units, quoted mass with plateau flag (pl = plateau). VR effective masses $m(1), m(2), m(3)$ in the last column.

β	lattice	N	ORIG	VR	CP	VR $m(1), m(2), m(3)$
2.3	8^4	5000	1.134(54) pl	0.719(83) late	0.863(24) pl	1.022(41), 0.719(83), 0.226(88)

β	lattice	N	ORIG	VR	CP	VR $m(1), m(2), m(3)$
2.3	12^4	3670	0.859(104) late	0.943(28) pl	0.863(19) pl	0.943(28), 0.944(93), 0.458(121)
2.3	16^4	1228	1.049(79) pl	0.691(103) late	0.847(26) pl	1.018(57), 0.691(103), 0.481(158)
2.3	$12^3 \times 32$	1510	1.216(42) pl	1.052(43) pl	0.880(20) pl	1.052(43), 1.004(126), 0.433(126)
2.3	$16^3 \times 32$	840	1.325(67) pl	0.891(104) late	0.926(26) pl	1.050(42), 0.891(104), 0.508(142)
2.4	12^4	3730	0.873(25) pl	0.680(18) pl	0.609(17) late	0.680(18), 0.688(34), 0.548(56)
2.4	16^4	1230	0.942(47) pl	0.669(22) pl	0.640(28) late	0.669(22), 0.661(51), 0.577(91)
2.5	16^4	920	0.709(36) pl	0.475(18) pl	0.419(19) late	0.475(18), 0.467(32), 0.509(56)
2.5	20^4	560	0.701(36) pl	0.448(21) pl	0.371(22) late	0.448(21), 0.416(34), 0.436(56)

Units $a\sqrt{\sigma_{\mathrm{ORIG}}}$ from the largest box at each coupling: 0.383(26) (2.3, 16^4), 0.350(18) (2.3, $16^3 \times 32$), 0.2789(34) (2.4), 0.2072(10) (2.5). Free two-gluon level $4\sin(\pi/L)$: 1.035 ($L = 12$), 0.780 (16), 0.626 (20).

Table 2. Volume tests (basis A) and ratios at the largest box.

β	boxes	ρ_{VR}	ρ_{ORIG}	ρ_{CP}	VR call	$m_{\mathrm{VR}}/m_{\mathrm{ORIG}}$	$m_{\mathrm{VR}}/\sqrt{\sigma}$
2.3	$16^4/12^4$	0.733(111)	1.22(18)	0.981(37)	gapless	0.66(10)	1.80(30)
2.3	$16^3 \times 32/12^3 \times 32$	0.847(105)	1.089(67)	1.053(38)	inconclusive	0.673(83)	
2.4	$16^4/12^4$	0.984(42)	1.079(62)	1.051(55)	gap	0.711(36)	2.40(8)
2.5	$20^4/16^4$	0.944(57)	0.989(72)	0.886(66)	gap	0.639(43)	2.16(10)

At $\beta = 2.3$ the VR effective mass falls with t on 8^4, 16^4 and $16^3 \times 32$ and holds only on the 12 boxes; the plateau rule then quotes $m(2)$ on one box and $m(1)$ on the other, which drives the “gapless” call of series 1 and the inconclusive call of series 2. The $\beta = 2.3$ VR channel is not under control. At $\beta = 2.4$ every VR effective mass plateaus. At $\beta = 2.5$ $m(1) \approx m(2) \approx m(3)$ in both boxes.

4.3 String tension and the premise check

E1 ratios $\sigma_{\mathrm{VR}}/\sigma_{\mathrm{ORIG}}$: $-0.09(4)$, $-0.04(1)$, $-0.02(1)$ ($\beta = 2.3$; $L = 8, 12, 16$), $-0.048(8)$, $-0.024(10)$ (2.3, $T = 32$), $-0.015(8)$, $+0.016(8)$ (2.4), but $0.219(10)$ (16^4) and $0.205(7)$ (20^4) at 2.5: C0 fails at $\beta = 2.5$. CP ratios are 0.94 to 1.17 everywhere (series 1 and 2).

Table 3. Long-time estimators on the series 2 configurations at $\beta = 2.5$ (single gauge copy), ratio VR/ORIG.

lattice	**E1**	**E1′ ($R \geq 4$)**	**E2, $T = 5$**	**E2, $T = 6$**	$\chi(5,5)$	$\chi(6,6)$	$\chi(7,7)$
16^4	0.219(10)	0.191(25)	−0.079(29)	−0.190(62)	0.429(24)	0.207(56)	0.062(101)
20^4	0.205(7)	0.213(13)	−0.046(21)	−0.029(52)	0.397(19)	0.237(50)	0.044(80)

The VR Creutz ratios at 16^4 fall $0.97, 0.83, 0.63, 0.43, 0.21, 0.06$ for $R = 2$ to 7; at $\beta = 2.3$ the same sequence is already 0.07 at $R = 4$. The VR potential at 2.5 rises out to $R \approx 6a$ and is flat within errors beyond, so the flattening distance moves outward in lattice units as a shrinks, roughly as for a fixed physical scale. E1 fits $T = 3 \to 4$ over $R = 2$ to 8 and is dominated by the rising intermediate-distance part. (The negative E2 value at $T = 6$ on 16^4 is a fit artifact of $V_0 + \sigma R - e/R$ with free e on a potential that is flat beyond $R \approx 6$, not a negative string tension.) This analysis was done after the C0 failure was known and is not blind.

4.4 Gauge-copy study at $\beta = 2.5$ (series 3)

Rules were fixed before the series 3 data existed. The best-of-16 functional exceeds the single copy on 94% of the 668 16^4 configurations, by 0.00215(5) on average (0.79115 → 0.79330; cf. [14]). The single-copy E1 ratio 0.205(9) reproduces series 2 (0.219(10)) on independent configurations, and the $\beta = 2.3$ control passes ($-0.046(21)$).

Table 4. Series 3, 16^4, $\beta = 2.5$, $N = 668$: string-tension ratios to ORIG, single copy vs best of 16 (paired difference).

estimator	**VR single**	**VR best of 16**	**VR diff.**	**CP single**	**CP best of 16**	**CP diff.**
E1	0.205(9)	0.225(10)	+0.020(13)	0.985(19)	0.726(14)	−0.259(12)
E2, $T = 5$	−0.042(37)	−0.044(37)	−0.002(50)	1.086(70)	0.765(50)	−0.321(28)
E2, $T = 6$	−0.045(95)	−0.151(109)	−0.106(135)	1.70(33)	1.16(22)	−0.54(11)
$\chi(6,6)$	0.232(72)	0.357(66)	+0.125(83)	1.013(94)	0.728(67)	−0.285(31)
$\chi(7,7)$	0.188(217)	0.106(200)	−0.082(285)	1.35(51)	0.93(35)	−0.42(16)

Ranking the 16 candidates by functional, the CP E1 ratio falls monotonically from 1.32 (worst) to 0.73 (best), while the VR E1 ratio rises mildly from 0.17 to 0.23. The VR glueball mass (basis A) moves from 0.452(23) (plateau) with the single copy to 0.561(44) (late) with the best copy, a paired difference of +0.109(48) ($m(1)$: +0.051(28)); the single-copy value agrees with series 2 (0.475(18)). The 20^4 sample (54 configurations) gives E1 VR ratios 0.223(17) and 0.252(23), CP 0.970(22) and 0.754(22); it is too small for masses.

5. Pre-registered outcomes

Each series fixed its combined rule in advance; we report them as written.

- Series 1: "gapless", triggered by $\beta = 2.3$ alone ($\rho_{\mathrm{VR}} = 0.733(111)$) while $\beta = 2.4$ was a clean "gap".
- Series 2: the combined rule as written reads "gap survives vortex removal (in this window)" ($\beta = 2.5$ gap, 2.4 gap, no significant drop of $m_{\mathrm{VR}}/\sqrt{\sigma}$ from 2.4 to 2.5, 2.3 not gapless). The rule did not say whether the series 1 consequence of a C0 failure (that coupling does not enter the verdict) carries over; with it, the outcome is inconclusive.
- Series 3 was designed to settle that by test. Its rule: if the best-of-16 VR E1 ratio is still ≥ 0.15, the combined outcome is inconclusive, because the $\beta = 2.5$ VR point is not a clean vortex-free measurement under C0. The ratio is 0.225(10), so the **combined outcome is inconclusive**. Of the pre-registered diagnoses, a gauge-fixing artifact is excluded (the residual does not fall with better copies); "measurement artifact of the short-time estimator" and "real residual" are not separated, because the long-range estimators at 16^4 have errors of about 0.2.

6. Discussion

The firm result is at $\beta = 2.4$: on an ensemble whose string tension is zero within errors, the vortex-removed 0^{++} channel has a volume-independent decay rate of about 0.7 of the original glueball mass, below the free two-gluon level of both boxes. At $\beta = 2.5$ the same picture holds for the mass, and the long-time string-tension estimators agree with [1, 2] that vortex removal removes the string tension, but the pre-registered short-time check does not pass, and the VR mass moves at the 2σ level with the gauge copy. The center-projected string tension depends strongly on the gauge copy (0.99 of the full one with a single copy, 0.73 with the best of 16, falling monotonically with the functional, as in [14, 15]); since the VR links are the original links with the center part divided out, the VR ensemble itself depends on the copy, and a VR mass should be quoted with that systematic. At $\beta = 2.3$ the VR correlator is not a clean exponential at the available times.

What would settle it: a premise check built on long-time or Creutz-ratio estimators fixed before new data, larger T at $\beta = 2.5$ and finer couplings, a third box per coupling, and a systematic gauge-copy (or Laplacian center gauge) study of the VR mass.

Data and code

Data tables (CSV) and the analysis and simulation code accompany this note in the same repository, with instructions to reproduce.

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Entries marked [recalled] were not opened in full text for this note; their bibliographic data were checked against arXiv metadata.

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